Jinglong Xu

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Education

- **Swiss Federal Institute of Technology Zürich (ETH Zürich)** — Switzerland
  Ph.D Candidate — *Oct. 2023 – Present*
  Advisor: *Prof. Hua Wang* ***Research Interests:*** *RF/mm-Wave Integrated Circuits and Systems, Phased Arrays, Power Amplifiers*
  Honors: *Winner of* ***EuMIC 2025*** *Young Engineer Prize*

- **Korea Advanced Institute of Science and Technology (KAIST)** — South Korea
  M.S. – Electrical Engineering; GPA: 91.33/100 — *Sep. 2020 – Aug. 2022*
  Advisor: *Prof. Sang-Gug Lee* ***M.S. Thesis Title:*** *Design of CMOS Receivers for 5G NR and LPWAN Applications*
  Honors: *Recipient of* ***ISSCC 2022*** *Student Travel Grant*

- **Harbin Institute of Technology (HIT)** — China
  B.E. – Material Science and Engineering; Major GPA: 3.76/4.00 — *Sep. 2016 – Aug. 2020*
  Honors: *Recipient of the National Scholarship by China Scholarship Council (****CSC****); First Prize of People's Scholarship (2016 & 2017)*

- **University College Dublin (UCD)** — Ireland
  CSC Outstanding Undergraduate Exchange Program; GPA: 3.98/4.20 — *Sep. 2018 – Aug. 2019*
  Major Courses: *Radio-Frequency Electronics (****A****+); Solid-State Electronics (****A****+); Biophysics at the Nanoscale and Nanodevices (****A****+); Nanomaterials (****A****+); Advanced Metals/Materials Processing (****A****+); Physics of Nanomaterials (****A****+)*

Work Experience

- **RF Technical Consultant – Point2 Technology Inc.** — South Korea
  D-Band (176 GHz) Wireline Transceiver SoC in 90-nm SiGe for an 800 Gb/s Link — *Sep. 2022 – Jun. 2023*
  - **D-Band Receiver**: Designed key receiver sub-blocks, including mixer, clock path, power divider, and output driver.
  - **RX Optimization**: Conducted RX full-chain optimization for PAM-4, including LNA, mixer, LO, and output driver.
  - **Tape-outs**: 90-nm SiGe BiCMOS *Nov. 2022* (High-Band); 90-nm SiGe BiCMOS *Mar. 2023* (Dual-Band)

- **Contract Researcher – KAIST** — South Korea
  Sub-6-GHz CMOS Wireless Receivers for IoT & 5G NR Applications — *Sep. 2022 – Oct. 2023*
  - **Low-Power WuRX**: Led the receiver design for a $0.2 - 1.0$ GHz radio SoC integrating RF, MODEM, and PMU unit.
  - **Duty-Cycling Scheme**: Implemented a novel two-step wake-up scheme for the sub-10-μW duty-cycled WuRX.
  - **Time-Modulated RX**: Designed the DLL and RX for a novel time-divided I/Q switched capacitor RX architecture.
  - **Tape-outs**: 55-nm CMOS *Oct. 2022*; 65-nm CMOS *May. 2023* (Time-Modulated RX); 55-nm CMOS *Sep. 2023*

Research Experience

- **RF/mm-Wave Integrated Circuits and Systems for Satellite and 6G Applications** — ETH Zürich
  Ph.D. Candidate — *Oct. 2023 – Present*
  - **Power Amplifiers**: Developing advanced power amplifier architectures for Ku/Ka-band SATCOM phased-array systems, including high power density designs, compact single-footprint Doherty architectures, and VSWR-resilient solutions.
  - **Phased Arrays**: Developing Ku/Ka-band transmitters for SATCOM arrays featuring VSWR-resilient PAs and integrated digital/PMIC SoC; Developing novel dynamic array architectures for joint communication and sensing.
  - **Taped-outs**: 22-nm SOI; Completed 10^+ tapeouts in GF22 FDX+ with both building-block IPs and array-level systems.

- **Multi-Band Radio SoC with a Sub-10-μW Wake-Up RX for IoT Applications** — KAIST
  M.S. Candidate — *Aug. 2021 – Oct. 2022*
  - **RX Circuit**: Designed LNA, mixer, baseband PGA, complex IRNF, and BFSK demodulator for the RX signal path.
  - **Duty-Cycling Scheme**: Proposed a novel two-step wake-up duty-cycling scheme for the sub-10-μW wake-up RX.
  - **Taped-out**: 55-nm CMOS; RX architecture verified, achieved -96 dBm sensitivity for 250 kb/s R_{chip} under 300 μW.

- **Sub-6-GHz Low Noise Front-End Module for 5G NR Applications** — KAIST
  M.S. Candidate — *Sep. 2020 – Aug. 2021*
  - **Ultra-Low-Noise LNA**: Designed an $n79$ $_{(4.7GHz)}$ gain-switching LNA with proposed linearity improvements.
  - **Testbench Modeling**: Developed a strong foundation in parasitics-aware RF design, with model-validated testbench.
  - **Taped-out**: 65-nm CMOS; Achieved the best FoM among LNAs at $4 - 6$ GHz, outperforming commercial benchmarks.

- **Graphite Nanoplatelet-Based Flexible Electronics and Thermal Management** — HIT
  Undergraduate Research Assistant — *Sep 2017 – Sep. 2020*
  - **Flexible Electronics**: Designed GNP-based flexible strain/pressure sensors with application on soft robotic control.
  - **Thermal Management**: Designed thermal interface structure for soft and high thermal conductivity PCB substrates.

Skills

- **Languages**: Mandarin (Native), English (Fluent, IELTS 7.5), Korean (Basic), German (Basic, Targeting B1).
- **IC Simulation**: Cadence Virtuoso, EMX, Keysight ADS, Matlab, Simulink, Ansys HFSS, and related EDA tools.
- **IC Evaluation**: Test Automation, NF and Sensitivity, Linearity, PA Power & Modulation, and related RF testing.
- **Soft Skills**: Strong Work Ethic, Communication, Teamwork, Leadership, Time Management, Public Speaking.
- **Hobbies**: Long-Distance Cycling, Bikepacking, Marathon Running, Skiing (Cross-Country & Alpine), Cooking.

Publications

Google Scholar ORCID

- **Journal**: **J. Xu**, E. Liu, M. Eleraky, T.-Y. Huang, C. Chu, H. Wang, "A Compact Ku-Band FR3 Doherty Power Amplifier with a Stacked Three-Coil Transformer Combiner: Leakage-Coupled Modeling and Schematic-to-EM Synthesis Methodology," ***TMTT 2026***. ***Manuscript In Preparation***
- **Journal**: **J. Xu**, T.-Y. Huang, M. Eleraky, I. Tolaib, B. Abdelmagid, H. Wang, "An Ultra-Compact Series Doherty Power Amplifier with a Single-Footprint Output Combiner for Ku-Band FR3," ***JSSC 2026 Special Issue***. ***Under Revision***
- **Conference**: **J. Xu**, T.-Y. Huang, M. Eleraky, I. Tolaib, H. Wang, "20.1 An Ultra-Compact Asymmetrically Load-Pulled Series Doherty PA in CMOS-SOI with 25.3dBm P_{sat} and 29.7% PAE_{6dB} for Ku-Band 6G FR3," ***ISSCC 2026***. ***Invited to JSSC 2026 Special Issue***
- **Conference**: **J. Xu**, E. Liu, M. Eleraky, T.-Y. Huang, C. Chu, H. Wang, "An Ultra-Compact Ku-Band Doherty PA with a Single-Footprint Triple Two-Turn Asymmetric Combiner for 6G FR3," ***EuMIC 2025***. ***Young Engineer Prize Winner***
- **Journal**: **J. Xu**, M. Eleraky, T.-Y. Huang, C. Chu, H. Wang, "A Compact Doubly Neutralized Ku Band Power Amplifier with 39% Peak PAE and 23 dBm Output Power in 22FDX+ EDMOS for 6G FR3," ***MWTL 2025***. ***Invited, IMS 2025 Top-50 Papers***
- **Conference**: **J. Xu**, K.-M. Kim, H.-U Mahmood, J. Kim and S.-G Lee, "An n79 Sub-1-dB Noise Figure Highly Linear Variable-Gain LNA Employing Adaptive Imbalanced Bleeding for 5G NR," ***A-SSCC 2022.***
- **Journal**: **J. Xu**, H. Wang, T. Ma, Y. Wu, R. Xue, H. Cui, X. Wu, Y. Wang, X. Huang, W. Yao, "A graphite nanoplatelet-based highly sensitive flexible strain sensor," ***Carbon 2020.*** ***IF: 11.6***

- **Journal**: D. Tran, H.-U Mahmood, **J. Xu**, K.-M. Kim, K.-S. Choi, J. Lee, J. Ko, S.-G. Lee, J. Kim, "A Multi-Band Ultra-Low-Power Transceiver with Two-Step Wake-Up Reception," ***SSCL 2026.***
- **Conference**: C. Chu, **J. Xu**, Y. Liu, J. Zeng, A. Wang, T. Torii, S. Shinjo, K. Yamanaka, H. Wang, "AI-Assisted Template-Seeded Pixelated Design for Multi-Metal-Layer High-Coupling EM Structures: A Ku-Band 6G FR3 PA in 22nm FDX+," ***IMS 2025.***
- **Conference**: D. Tran, H.-U Mahmood, **J. Xu**, K.-M. Kim, K.-S. Choi, J. Lee, J. Ko, S.-G. Lee, J. Kim. "Multi-band LPWAN Transceiver with Two-step Wake-up Receiver and Switched Capacitor PA Based Transmitter Achieving -125 dBm Sensitivity, 6.7 uW at 1 s Wake-up Latency, and 1Mbps BFSK Modulation," ***A-SSCC 2025.***
- **Conference**: H.-U Mahmood, K.-M. Kim, D. Tran, **J. Xu**, A. Qahir, J. Ko, J. Kim, S.-G. Lee, K.-S. Choi, "A 24% Efficient, 15.36 dBm Output Power, Multi-Standard Digital Polar Transmitter with 7-bit Phase Interpolator-based BFSK Modulator and 23 dB Sidelobe Suppressed PA for Low-Power Wide Area Networks," ***A-SSCC 2024.***
- **Journal**: K.-M. Kim, K.-S. Choi, H. Jung, B. Yun, **J. Xu**, J. Ko, S.-G. Lee, "A −124-dBm Sensitivity Interference-Resilient Direct-Conversion Duty-Cycled Wake-Up Receiver," ***JSSC 2023.***
- **Patent**: S.-G. Lee, K.-M. Kim, J. Ko, K.-S. Choi, **J. Xu**, E.-R. Jeong, "A Duty-Cycling Method for Wake-Up Radios," Korean Domestic Patent Number: 10-2022-0054080, 02 May, 2022.
- **Journal**: X. Wu, H. Wang, Z. Wang, **J. Xu**, Y. Wu, R. Xue, H. Cui, C. Tian, X. Huang, B. Zhong, "Highly Conductive Thermal Interface Materials with Vertically Aligned Graphite-Nanoplatelet Filler towards: High Power Density Electronic Device Cooling," ***Carbon 2021.*** ***IF: 11.6***
- **Journal**: Y. Du, Y. Xu, W. Zhou, Y. Yu, X. Ma, F. Liu, **J. Xu**, Y. Zhu, "MOF-derived zinc manganese oxide nanosheets with valence-controllable composition for high-performance Li storage," ***Green Energy & Environment 2021.*** ***IF: 10.7***
- **Journal**: X. Wu, Y. Wang, H. Wang, B. Zheng, Y. Wu, R. Xue, **J. Xu**, H. Cui, W. Yao, B. Zhong, X. Huang, "Semi-quantitative orientation control of graphite nanoplatelets in GNP/PU nanocomposite via balancing the effects of gravity and micro-flow field and application in manufacturing heat spreader substrate with excellent thermal conductivity," ***Composites Part A 2021.*** ***IF: 8.1***
- **Journal**: Y. Wang, H. Wang, F. Liu, X. Wu, **J. Xu**, H. Cui, Y. Wu, R. Xue, C. Tian, B. Zheng, W. Yao, "Flexible printed circuit board based on graphene/polyimide composites with excellent thermal conductivity and sandwich structure," ***Composites Part A 2021.*** ***IF: 8.1***

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